

CLOUD ENGINEERING · PROJECT 1

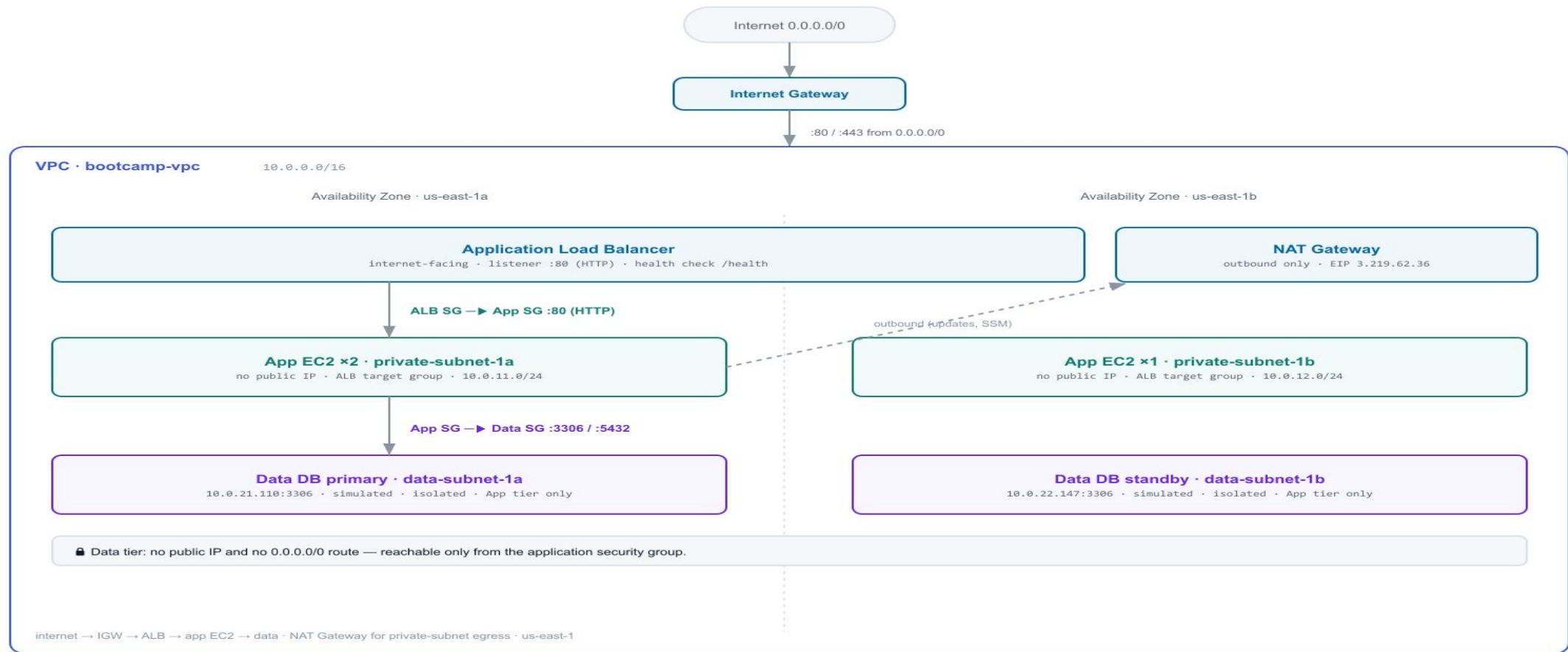
3-Tier Cloud Architecture on AWS

Designed, built, secured, tested, and cost-analyzed — presented through the live AWS console.

Architecture at a glance

OVERVIEW

3-Tier Architecture — bootcamp-vpc



VPC & subnets — 6 subnets across 2 AZs

The screenshot displays the AWS Management Console for a VPC named 'bootcamp-vpc' (ID: vpc-0917ed6d87b005f58). The interface is divided into a left-hand navigation pane and a main content area.

Left-hand navigation pane:

- VPC dashboard** (selected)
- Virtual private cloud**
 - Your VPCs
 - Subnets
 - Route tables
 - Internet gateways
 - Egress-only internet gateways
 - Carrier gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security**
 - Network ACLs
 - Security groups
- PrivateLink and Lattice**
 - Getting started
 - Endpoints
 - Endpoint services
 - Service networks
 - Lattice services
 - Resource configurations
 - Resource gateways
 - Target groups
 - Domain

Main content area:

vpc-0917ed6d87b005f58 / bootcamp-vpc

Details

Property	Value
VPC ID	vpc-0917ed6d87b005f58
State	Available
Block Public Access	Off
DNS resolution	Enabled
Tenancy	default
DHCP option set	dopt-0a5c577c1bb52b873
Main network ACL	acl-02eb9cc3b687c1217
Default VPC	No
IPv4 CIDR (Network border group)	10.0.0.0/16
Network Address Usage metrics	Disabled
Route 53 Resolver DNS Firewall rule groups	-
Encryption control ID	-
Encryption control mode	-
DNS hostnames	Enabled
Main route table	rtb-077bc323c4fe62757
IPv6 pool	-
Owner ID	697345203222

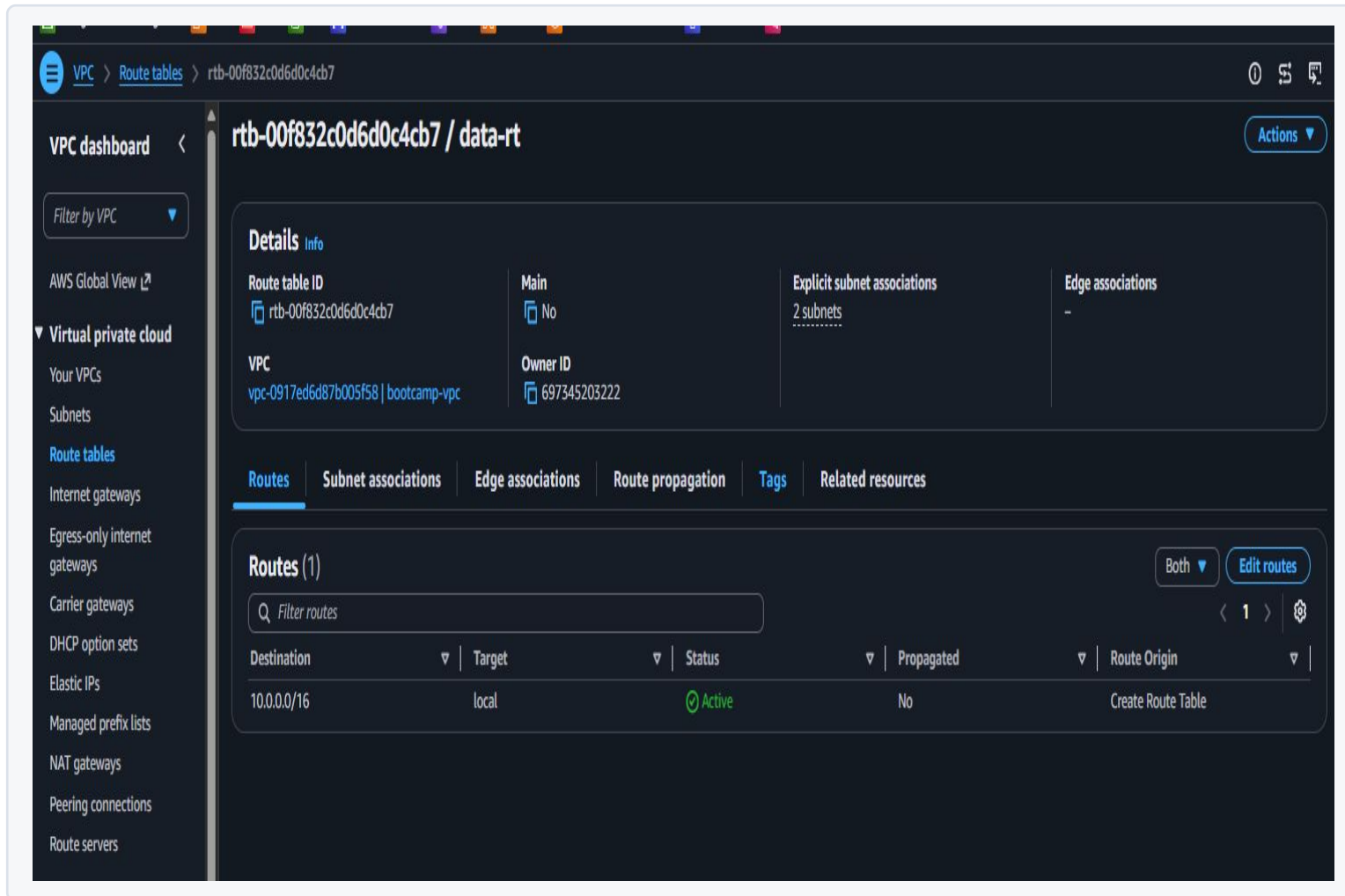
Resource map

The resource map shows the following components:

- VPC:** bootcamp-vpc
- Subnets (6):**
 - us-east-1a:** public-subnet-1a, data-subnet-1a, private-subnet-1a
 - us-east-1b:** public-subnet-1b, private-subnet-1b, data-subnet-1b
- Route tables (4):** app-rt, data-rt, public-rt, rtb-077bc323c4fe62757
- Network Connections:** bootcamp-igw, bootcamp-nat-1a

- One VPC, 10.0.0.0/16
- 2 public + 2 app + 2 data subnets
- Spread across us-east-1a and 1b
- /24 each — room to grow

Data-tier isolation — route table



- data-rt has only the local route
- No 0.0.0.0/0 → no path to the internet
- Isolated by routing, not just firewall

The three tier security groups

EC2 > Security Groups

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
data-tier-sg	sg-01c6914f0ca937cb1	data-tier-sg-3tier	vpc-0917ed6d87b005f58	Tier3 Data - MySQL/PostgreSQL from
app-tier-sg	sg-0b8d55053c76f6e66	app-tier-sg-3tier	vpc-0917ed6d87b005f58	Tier2 App - HTTP from ALB only
-	sg-03a815ee5f22d41b9	default	vpc-0917ed6d87b005f58	default VPC security group
alb-sg	sg-0afa31c9844b6cea0	alb-sg-3tier	vpc-0917ed6d87b005f58	Tier1 ALB - HTTP/HTTPS from interne
-	sg-0a00e8e52f14fc969	default	vpc-05e8c5246cd31a701	default VPC security group

- ALB → App → Data chain
- One SG per tier
- Each trusts only the tier in front

Least privilege — App SG inbound

The screenshot shows the AWS Management Console interface for the security group **sg-0b8d55053c76f6e66 - app-tier-sg-3tier**. The console is in the **EC2** service, under **Security Groups**.

Details:

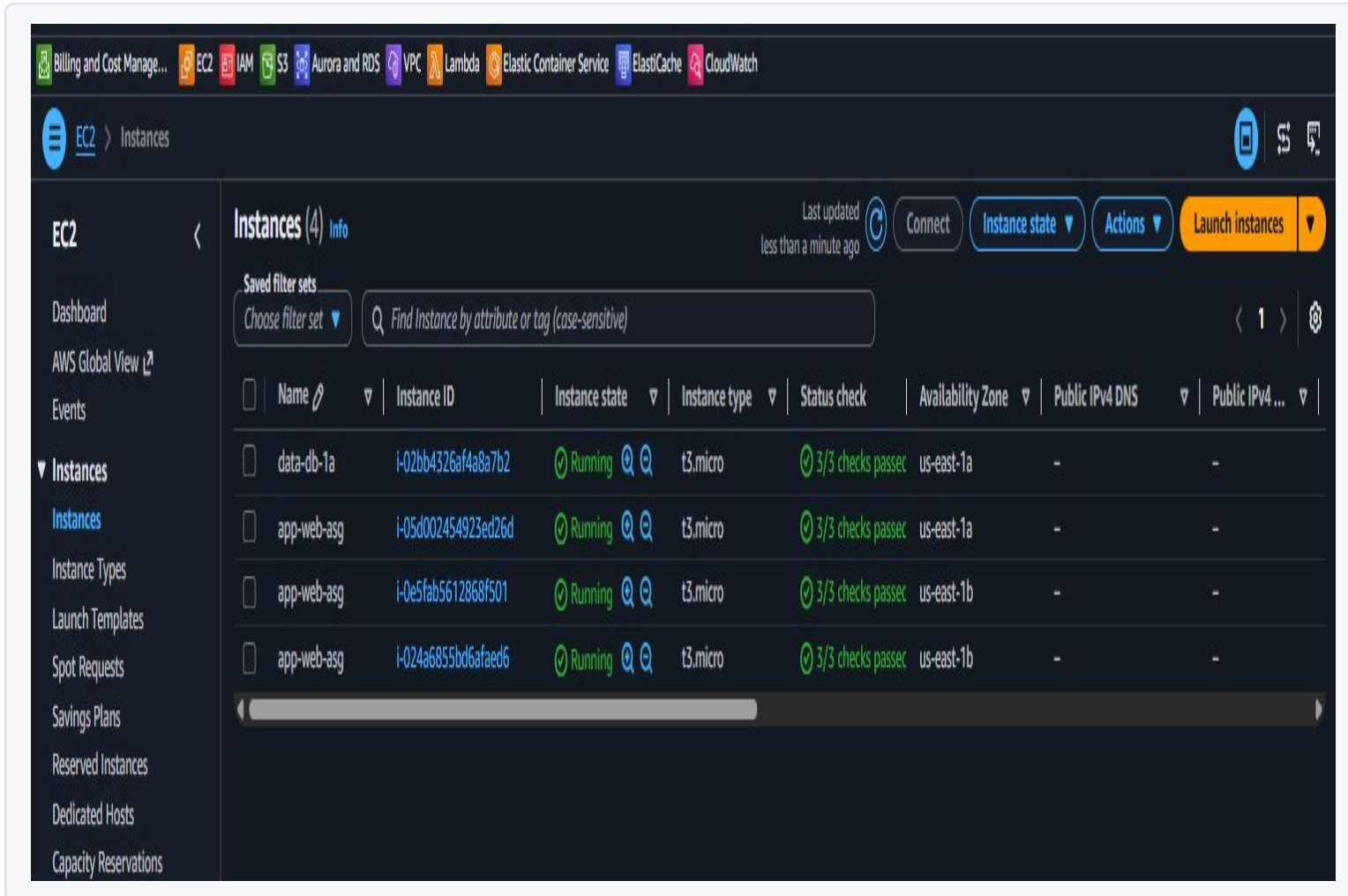
Security group name	app-tier-sg-3tier	Security group ID	sg-0b8d55053c76f6e66	Description	Tier2 App - HTTP from ALB only	VPC ID	vpc-0917c06d87b0c9f58
Owner	697345203222	Inbound rules count	1 Permission entry	Outbound rules count	1 Permission entry		

Inbound rules (1):

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0c54c0ffc65d8a07a	-	HTTP	TCP	80	sg-0efa311c9844b6ee0	HTTP from ALB

- App SG: :80 from ALB SG only
- SG-to-SG reference, not IP ranges based on identity not address
- Direct hits to instances are dropped if not coming from ALB security Group

EC2 instances across both AZs



The screenshot displays the AWS Management Console for EC2 instances. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of four instances, all in a 'Running' state. The instances are distributed across two availability zones: us-east-1a and us-east-1b. The table below summarizes the instance details.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
data-db-1a	i-02bb4326af4a8a7b2	Running	t3.micro	3/3 checks passed	us-east-1a	-	-
app-web-asg	i-05d002454923ed26d	Running	t3.micro	3/3 checks passed	us-east-1a	-	-
app-web-asg	i-0e5fab5612868f501	Running	t3.micro	3/3 checks passed	us-east-1b	-	-
app-web-asg	i-024a6855bd6afaed6	Running	t3.micro	3/3 checks passed	us-east-1b	-	-

- App instances + data nodes running
- Private subnets, no public IP
- Spread over us-east-1a / 1b

Application Load Balancer

The screenshot displays the AWS Management Console interface for an Application Load Balancer (ALB) named 'ce-app-alb'. The left sidebar shows the navigation menu with categories like EC2, Instances, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the details of the selected ALB, including its status (Active), VPC, Availability Zones, and DNS name. Below the details, there are tabs for Listeners and rules, Network mapping, Resource map, Security, Monitoring, Integrations, Attributes, Capacity, and Tags. The 'Listeners and rules' tab is active, showing a table of listeners and rules.

ce-app-alb

Details

Load balancer type Application	Status Active	VPC vpc-0917ed6d87b005f58	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0c28b42a4bdc84fda us-east-1b (use1-az4) subnet-0a8a1e9afe3d8117b us-east-1a (use1-az2)	Date created July 23, 2026, 14:32 (UTC+02:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:697345203222:loadbalancer/app/ce-app-alb/f20fc688cc80a4e6		DNS name ce-app-alb-1960431634.us-east-1.elb.amazonaws.com (A Record)	

Listeners and rules

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:80	Forward to target group ce-app-tg (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable
HTTPS:443	Forward to target group ce-app-tg (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-2016-08	ce-app-alb-1960431634.us-east-1.elb.amazonaws.com	Off	Not applicable

- Internet-facing
- Listeners :80 (HTTP) and :443 (HTTPS)
- Spans both Availability Zones

Target group — 3/3 healthy

The screenshot displays the AWS Management Console for the 'ce-app-tg' target group. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The 'Load Balancing' section is highlighted, and 'Target Groups' is selected.

Details

arn:aws:elasticloadbalancing:us-east-1:697345203222:targetgroup/ce-app-tg/3f16f11f811aa77e

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-0917ed6d87b005f58
IP address type IPv4	Load balancer ce-app-alb		

5 Total targets 3 Healthy 0 Unhealthy 0 Unused 0 Initial 2 Draining

0 Anomalous

► Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets Monitoring Health checks Attributes Tags

Registered targets (5) [Info](#) Anomaly mitigation: Not applicable Deregister Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...	Launch...
☐	i-0283bd36486f86548	app-web-asg	80	us-east-1a (us...)	Healthy	-	No override	No overri...	July 23, 2...
☐	i-0e2cf92a075f9d946	app-web-asg	80	us-east-1b (us...)	Healthy	-	No override	No overri...	July 23, 2...
☒	i-024a6855bd6afaed6	app-web-asg	80	us-east-1b (us...)	Draining	Target deregistration i...	No override	No overri...	July 23, 2...
☐	i-05d002454923ed26d	app-web-asg	80	us-east-1a (us...)	Healthy	-	No override	No overri...	July 23, 2...
☒	i-0e5fab5612868f501	app-web-asg	80	us-east-1b (us...)	Draining	Target deregistration i...	No override	No overri...	July 23, 2...

- All targets healthy
- Health check: GET /health
- Live load-balancing: [app-load-balancing.gif](#) (repo)

Auto Scaling Group

The screenshot displays the AWS Management Console interface for an Auto Scaling Group (ASG) named 'ce-app-asg'. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area is divided into several sections:

- ce-app-asg Capacity overview:** Shows the desired capacity (3), scaling limits (3 - 4), desired capacity type (Units (number of instances)), and status (At desired capacity). It also indicates the date created (Thu Jul 23 2026 16:12:09 GMT+0200).
- Details tab:** Contains sub-sections for:
 - Launch template:** Details the launch template (lt-084477dff93c22b43), AMI ID (ami-0f303bae6b70e0ed), instance type (t3.micro), security groups (sg-0b8d55053c76f6e66), and key pair name.
 - Network:** Shows availability zones (use1-az4, use1-az2), subnet ID (subnet-049f32c8e8e534e85), and availability zone distribution (Balanced best effort).
 - Instance type requirements:** States that the ASG adheres to the launch template for purchase option and instance type.
 - Health checks:** Shows the health check type (EC2, ELB) and health check grace period (120).

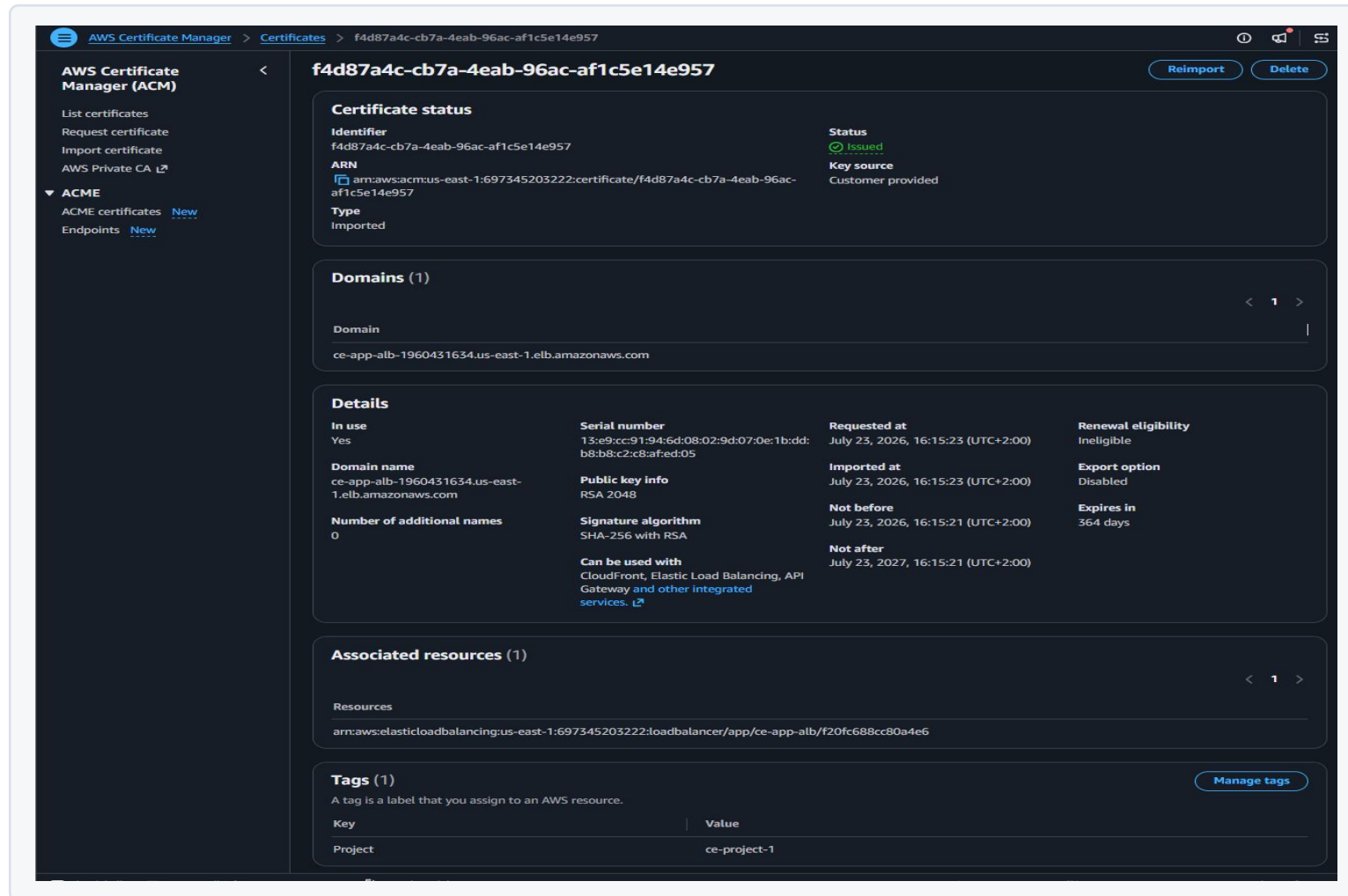
- min 2 / desired 3 / max 4
- Replaces failed instances automatically
- Instances across 2 AZs

CPU target-tracking policy

The screenshot displays the AWS Management Console interface for an Auto Scaling group named 'ce-app-asg'. The left-hand navigation pane shows the 'EC2' section expanded, with 'Auto Scaling Groups' selected. The main content area is titled 'ce-app-asg' and includes a 'Capacity overview' section showing a desired capacity of 3, scaling limits of 3-4, and a status of 'Updating capacity'. Below this, the 'Automatic scaling' tab is active, showing a notification about scaling policies. The 'Dynamic scaling policies' section lists one policy, 'ce-cpu-tracking', which is a 'Target tracking scaling' policy. This policy is 'Enabled' and configured to 'Execute policy when' 'As required to maintain Average CPU utilization at 50'. The 'Take the action' is 'Add or remove capacity units as required', and the 'Instances need' '300 seconds to warm up before including in metric'. The 'Scale in' is 'Enabled'. At the bottom, the 'Predictive scaling policies' section is empty, with a message stating 'No predictive scaling policies have been created'.

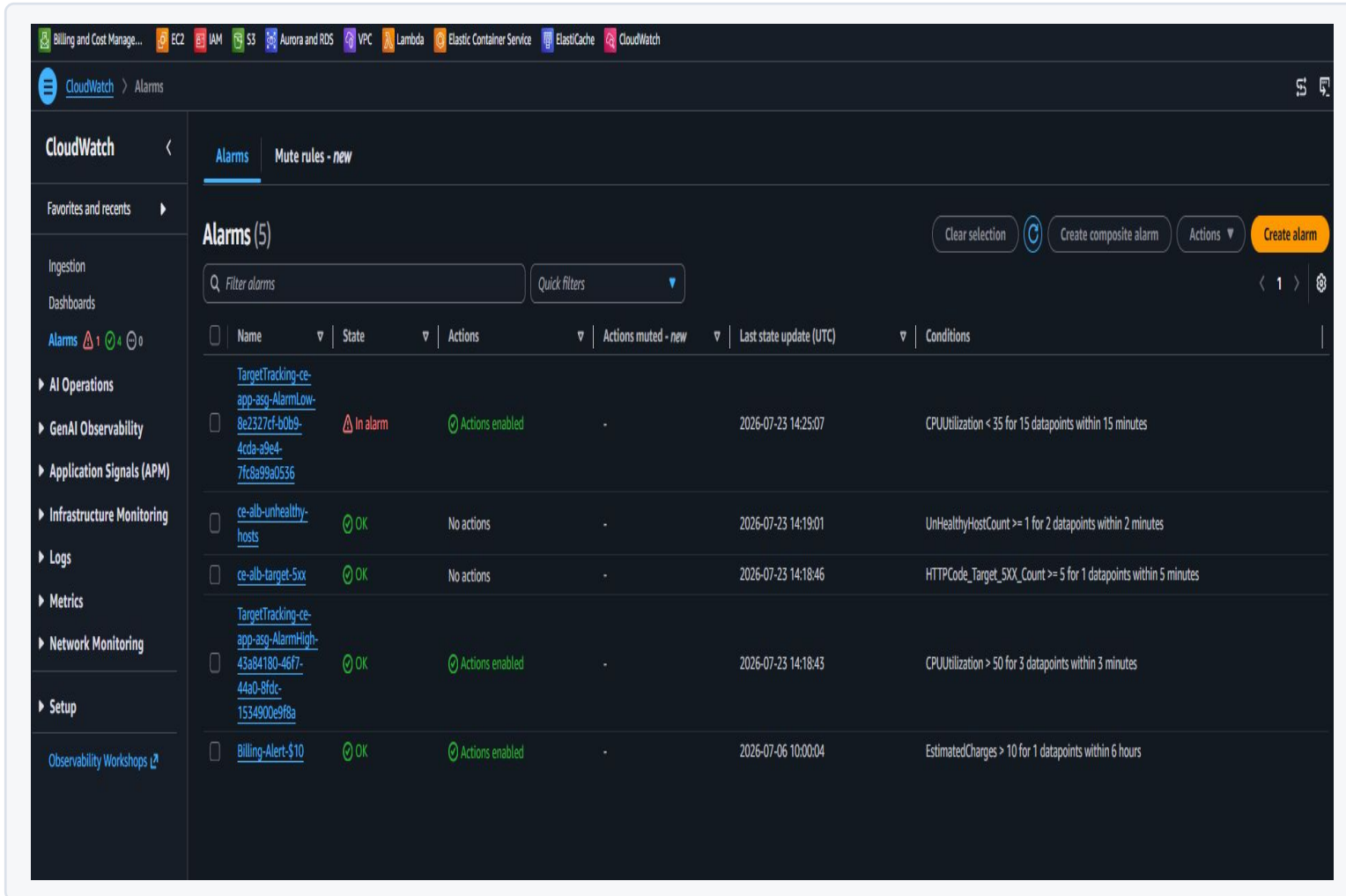
- Scales on average CPU = 50%
- Scales out to 4 under load
- Floor of 3 guarantees the HA baseline

HTTPS — ACM certificate



- TLS on the ALB (:443)
- ACM-managed certificate
- Self-signed for the demo

CloudWatch alarms



The screenshot displays the AWS CloudWatch Alarms console. The left sidebar shows the navigation menu with 'Alarms' selected. The main area shows a list of 5 alarms. The first alarm, 'TargetTracking-ce-app-asp-AlarmLow-8e2327cf-b0b9-4cda-a9e4-7fc8a99a0536', is in an 'In alarm' state. The other four alarms are in an 'OK' state. The table columns are: Name, State, Actions, Actions muted - new, Last state update (UTC), and Conditions.

Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
TargetTracking-ce-app-asp-AlarmLow-8e2327cf-b0b9-4cda-a9e4-7fc8a99a0536	In alarm	Actions enabled	-	2026-07-23 14:25:07	CPUUtilization < 35 for 15 datapoints within 15 minutes
ce-alb-unhealthy-hosts	OK	No actions	-	2026-07-23 14:19:01	UnHealthyHostCount >= 1 for 2 datapoints within 2 minutes
ce-alb-target-5xx	OK	No actions	-	2026-07-23 14:18:46	HTTPCode_Target_5XX_Count >= 5 for 1 datapoints within 5 minutes
TargetTracking-ce-app-asp-AlarmHigh-43a84180-46f7-44a0-8fde-1534900e9f8a	OK	Actions enabled	-	2026-07-23 14:18:43	CPUUtilization > 50 for 3 datapoints within 3 minutes
Billing-Alert-\$10	OK	Actions enabled	-	2026-07-06 10:00:04	EstimatedCharges > 10 for 1 datapoints within 6 hours

- Unhealthy hosts + target 5XX
- Alerting on the ALB health
- Plus billing guardrail

VPC Flow Logs

The screenshot displays the AWS Management Console interface for a VPC named 'bootcamp-vpc' (ID: vpc-0917ed6d87b005f58). The left sidebar shows the navigation menu with categories like 'Virtual private cloud', 'Security', and 'PrivateLink and Lattice'. The main content area is divided into a 'Details' section and a 'Flow logs' section.

VPC Details:

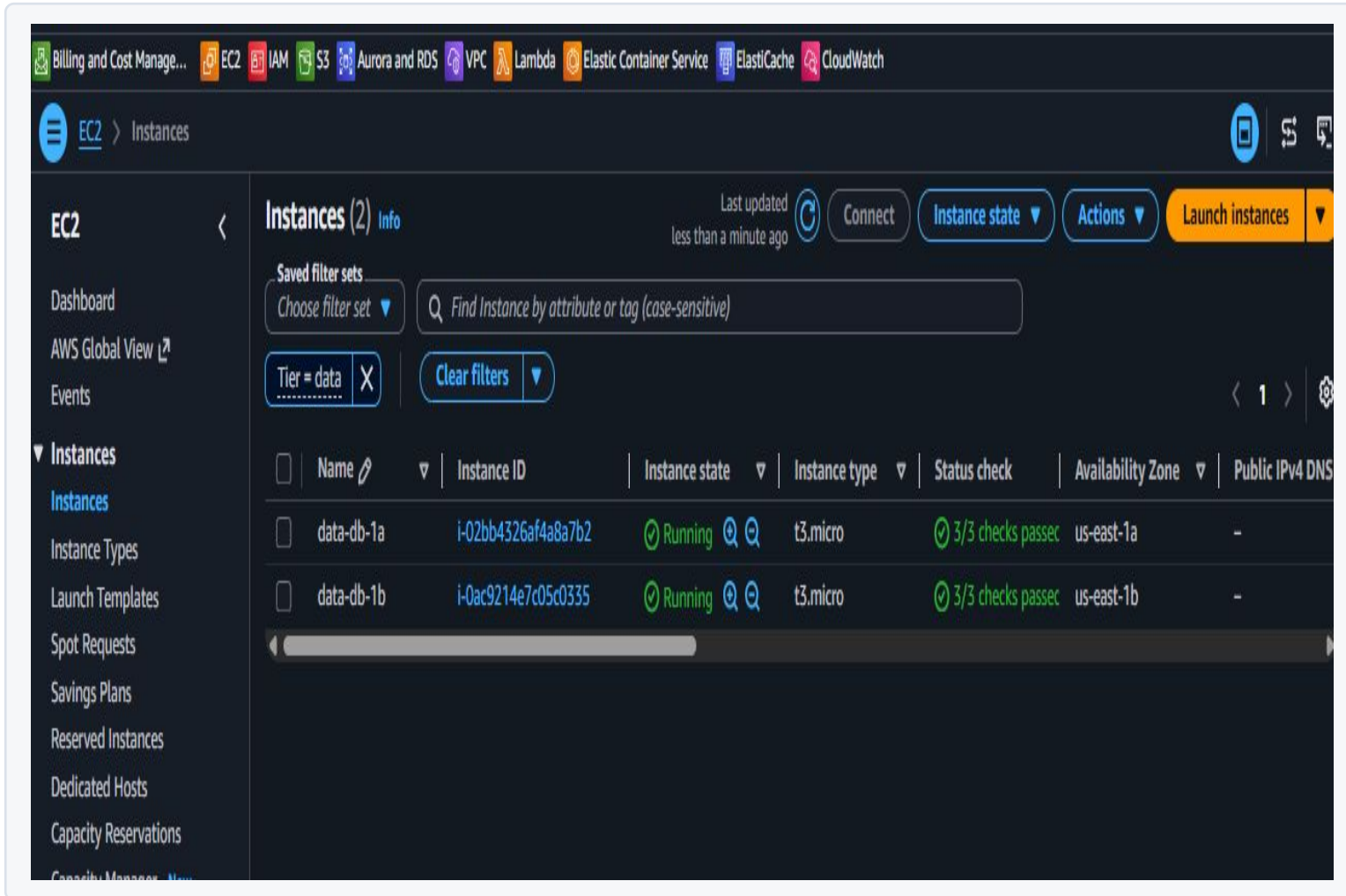
- VPC ID:** vpc-0917ed6d87b005f58
- State:** Available
- Block Public Access:** Off
- DNS hostnames:** Enabled
- DNS resolution:** Enabled
- Tenancy:** default
- DHCP option set:** dhcp-3a5c577c1b652b673
- Main route table:** rtb-077b323c4e62757
- Main network ACL:** acl-02eb9cc3b687c1217
- Default VPC:** No
- IPv4 CIDR:** 10.0.0.0/16
- IPv6 pool:** -
- IPv6 CIDR (Network border group):** -
- Network Address Usage metrics:** Disabled
- Route 53 Resolver DNS Firewall rule groups:** -
- Owner ID:** 697345203222
- Encryption control ID:** -
- Encryption control mode:** -

Flow logs (1):

Flow log ID	Traffic type	Destination type	Destination name	IAM role ARN	Max...	Crea...	Status
fl-832b57dea34896365	All	cloud-watch-logs	/vpc-re-project-1/flowlogs-1	arn:aws:iam::697345203222:role/vp-...	10 min...	Thursda...	Active

- All VPC traffic → CloudWatch
- Network forensics & troubleshooting
- Enabled at the VPC level
- records details about IP traffic going to and from network interfaces in a Virtual Private Cloud

Multi-AZ data — primary + standby

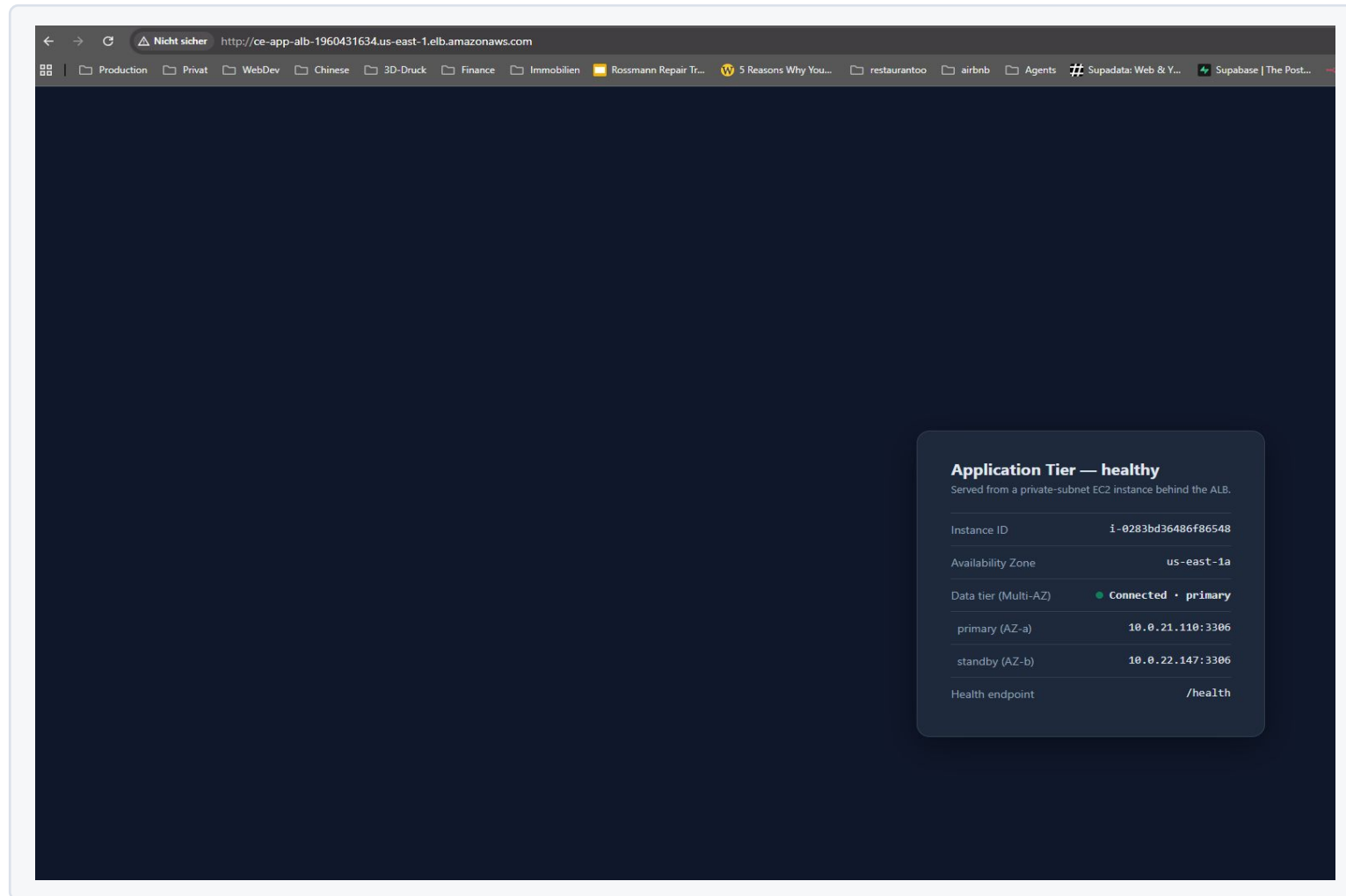


The screenshot displays the AWS Management Console interface for EC2 instances. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of two EC2 instances, 'data-db-1a' and 'data-db-1b', both in the 'Running' state. The instances are t3.micro type, located in us-east-1a and us-east-1b availability zones respectively. The status check for both instances shows '3/3 checks passed'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
data-db-1a	i-02bb4326af4a8a7b2	Running	t3.micro	3/3 checks passed	us-east-1a	-
data-db-1b	i-0ac9214e7c05c0335	Running	t3.micro	3/3 checks passed	us-east-1b	-

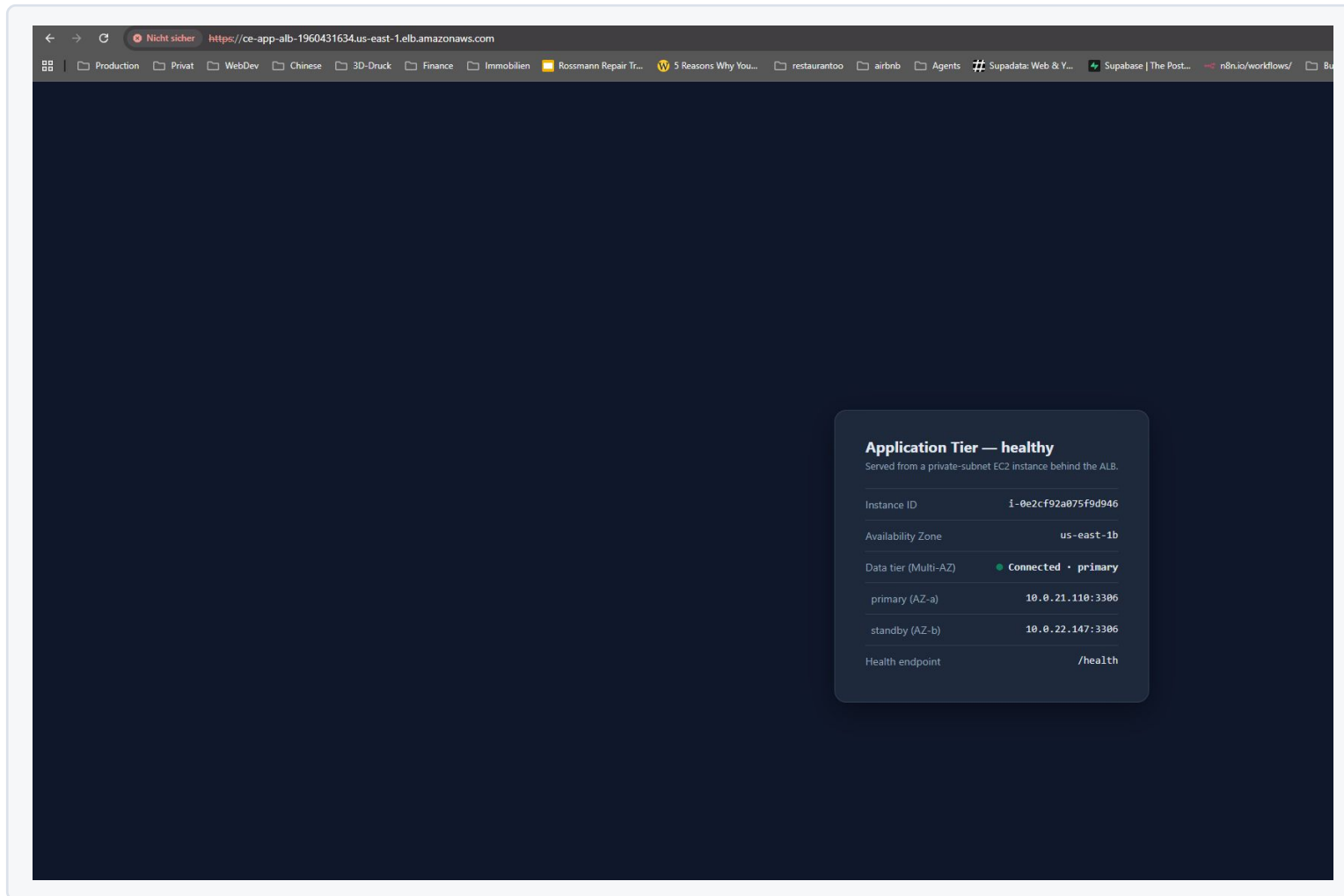
- primary in 1a, standby in 1b
- App fails over primary → standby
- Isolated subnets, app-tier access only

Running app over HTTP



- Instance-id + AZ shown
- Data tier: Connected (Multi-AZ)
- Refresh = different instance (balanced)

Running app over HTTPS



- Same app over TLS :443
- Cert warning expected (self-signed)

Verified: balanced, self-healing, Multi-AZ

3/3

targets healthy

6/6/6

even load spread

~30s

auto-recovery

failover

primary→standby

- Load: 18 requests spread evenly across all 3 instances and both AZs
- App-tier failover: stop an instance → ALB(ApplicationLoadBalancer) routes around it, ASG(AutoScalingGroup) relaunches it
- Data-tier failover: stop the primary → app flips to the standby, stays up

COST ANALYSIS

~\$101/month running — \$0 when torn down

~\$35

NAT Gateway

~\$22

Load Balancer

~\$30

EC2 (ASG+data)

\$0

after teardown

- NAT + ALB are ~57% of the bill and run regardless of traffic
- Biggest lever = don't leave it running idle (stop/start script, or full teardown)
- Spot Instance for the app tier — ~70% off compute

What broke, and how I fixed it

- Blackholed NAT route (going nowhere) — reused VPC pointed at a deleted NAT → new NAT and pointed it at the route
- Data-tier isolation — split the data subnets onto their own route table, no internet route
- Scaling rule removed too many instances — raised the ASG(Auto Scaling Group) floor to min 3 to hold the HA(High Availability) baseline because rule removed instances to save money (CPU was below 50% because nobody used it)

Path to production

Done as bonus

- Auto Scaling Group + CPU scaling
- HTTPS / ACM (self signed cert)
- VPC Flow Logs + alarms
- Multi-AZ data tier

Next

- Multi-AZ data → swap simulated DB(python script) for managed RDS
- HTTPS → public Amazon signed ACM cert + custom domain
- Infrastructure as Code (Terraform)
- CI/CD · WAF(Web Application Firewall) · multi-region DR(Disaster Recovery)

THANK YOU

Questions?

A production-shaped 3-tier architecture: multi-AZ HA, least-privilege security, auto-scaling, HTTPS, monitoring, and a Multi-AZ data tier — all evidenced in the AWS console.